

Extending not-MIWAE for correlated missing mechanisms

Mathéo Kina ¹ Yee-Yang Hsieh ¹

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Background

Problem: Real-world datasets often exhibit complex **Missing Not At Random** (MNAR) patterns with **correlated missingness across features**. While **not-MIWAE** (Ipsen et al. 2020) incorporates information about the missing mechanism, it assumes conditional independence of missingness given the data. Our goal is to improve imputation performance in the presence of correlated missing mechanisms.

Notation. We consider a dataset $\mathbf{X} = (x_1, \dots, x_n)^\top \in \mathcal{X}^n$, where each sample $x_i \in \mathbb{R}^p$ consists of observed features x_i^o and missing features x_i^m . Missingness is described by a mask $\mathbf{s} \in \{0, 1\}^p$. The joint distribution over data and missingness is factorized as $p_{\theta, \varphi}(x_i, s_i) = p_{\theta}(x_i) p_{\varphi}(s_i | x_i)$.

MNAR: Missingness depends on both observed and missing features, $p_{\varphi}(s_i | x_i) = p_{\varphi}(s_i | x_i^o, x_i^m)$.

The not-MIWAE Model

The model consists of a latent variable $z \sim p(z)$ generating data via $p_{\theta}(x|z)$, and a missing mechanism $p_{\varphi}(s|x)$ predicting masks from complete data.

Since $p_{\theta}(x|z) = \prod_j p_{\theta}(x_j|z)$ factorizes, the log-joint likelihood is

$$\log p_{\theta, \varphi}(x^o, s) = \log \int \int p_{\varphi}(s|x^o, x^m) p_{\theta}(x^o|z) p_{\theta}(x^m|z) p(z) dz dx^m$$

Since this double integral is intractable, we introduce an inference network $q_{\gamma}(z|x^o)$ as an importance sampling proposal, yielding:

$$\log \mathbb{E}_{z \sim q_{\gamma}(z|x^o), x^m \sim p_{\theta}(x^m|z)} \left[\frac{p_{\varphi}(s|x^o, x^m) p_{\theta}(x^o|z) p(z)}{q_{\gamma}(z|x^o)} \right]$$

Following the IWAE framework, we approximate this expectation using K Monte Carlo samples:

$$\mathcal{L}_K(\theta, \varphi, \gamma) = \sum_{i=1}^n \mathbb{E} \left[\log \frac{1}{K} \sum_{k=1}^K w_{ki} \right]$$

where

$$w_{ki} = \frac{p_{\varphi}(s_i|x_i^o, x_{ki}^m) p_{\theta}(x_i^o|z_{ki}) p(z_{ki})}{q_{\gamma}(z_{ki}|x_i^o)}$$

and samples (z_{ki}, x_{ki}^m) are drawn from $q_{\gamma}(z|x_i^o) p_{\theta}(x^m|z)$.

Conditional Independence Assumption

Standard assumption in not-MIWAE:

$$p_{\varphi}(s|x) = \prod_{j=1}^p \text{Bern}(s_j | \pi_{\varphi, j}(x))$$

Limitations:

- Cannot model correlations beyond those induced by data values
- Ignores latent regimes or external factors affecting multiple features
- Example: survey fatigue, system failures, block-structured missingness

Our Extension: Latent Regime Variable

New latent variable: $u \sim \mathcal{N}(0, I)$ captures structured dependencies

$$p_{\theta, \varphi}(x, s, z, u) = p(z) p(u) p_{\theta}(x|z) p_{\varphi}(s|x, u)$$

Missing predictor:

$$p_{\varphi}(s|x, u) = \prod_{j=1}^p \text{Bern}(s_j | \pi_{\varphi, j}(x, u))$$

Shared u captures correlations between mask variables.

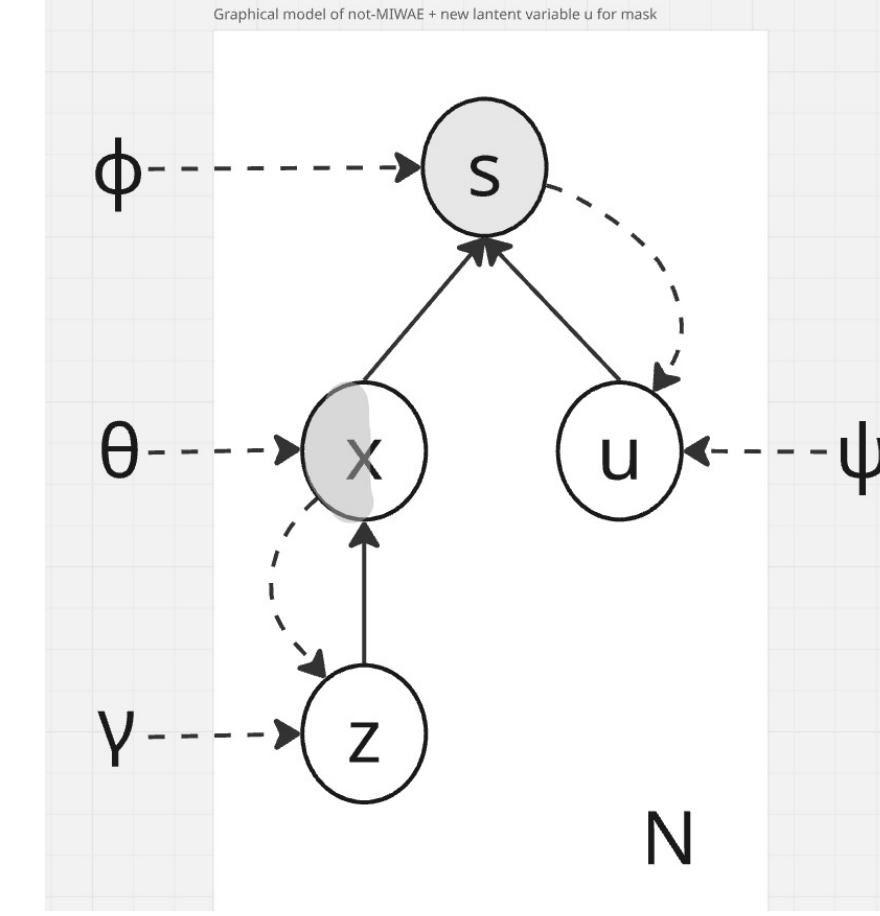


Figure 2. Extended model with regime u

Inference Framework

Two encoder approaches:

Full: $q_{\psi}(u|x^o, s) = \mathcal{N}(u | \mu_{\psi}(x^o, s), \sigma_{\psi}(x^o, s))$

Mask-only: $q_{\psi}(u|s) = \mathcal{N}(u | \mu_{\psi}(s), \sigma_{\psi}(s))$

Importance weights:

$$w_{ki} = \frac{p_{\varphi}(s_i|x_i^o, x_{ki}^m, u_{ki}) p_{\theta}(x_i^o|z_{ki}) p(z_{ki}) p(u_{ki})}{q_{\gamma}(z_{ki}|x_i^o) q_{\psi}(u_{ki}|x_i^o, s_i)}$$

Architecture Choices

Concatenation: $\pi([x, u]) = \sigma(\text{MLP}([x, u]))$

Additive: $\pi(x, u) = \sigma(\text{MLP}_x(x) + \text{MLP}_u(u))$

Multiplicative: $\pi(x, u) = \sigma(\text{MLP}_x(x) \times \text{MLP}_u(u))$

Experimental Setup

Missing mechanism: Two-level MNAR

- Self-censoring:** feature d missing if above mean
- Propagation:** feature $d + D/2$ also missing

Datasets: 5 UCI datasets

Training settings:

Table 1. Parameters used for main results

Parameter	Value
Training iterations	100,000
Importance samples (K)	20
Batch size	16
Data latent dimension	$D - 1$
Regime latent dimension (extended model)	10
Hidden layer size	128
Imputation samples (L)	5,000

Table 2. Missing rates for each dataset (%)

Dataset	Missing Rate (%)
Banknote	51.35
Concrete	41.88
Red Wine	39.02
White Wine	39.39
Breast	39.86

Main Results

Our extended model (full encoder) achieves lower imputation error than not-MIWAE across all five UCI datasets.

The extended not-MIWAE with the concatenation architecture achieves the best results on four of the five datasets.

Model	Bank	Concr.	Red	White	Breast
MIWAE	1.46	1.47	1.44	1.40	1.20
not-MIWAE	0.81	1.27	1.19	1.24	1.01
Ext. (concat)	0.76	1.19	1.18	1.20	1.04
Ext. (add.)	1.20	1.71	1.75	1.61	0.92
Ext. (mult.)	1.84	1.68	1.95	2.43	0.93
missForest	1.24	1.37	1.36	1.35	1.20

Table 3. Imputation RMSE across UCI datasets. Bold indicates the best model for each dataset.

Mask-only encoder fails to improve imputation. **Regime needs data access.**

Training Dynamics

Extended not-MIWAE requires high training iterations ($\geq 50,000$) to outperform the baseline.

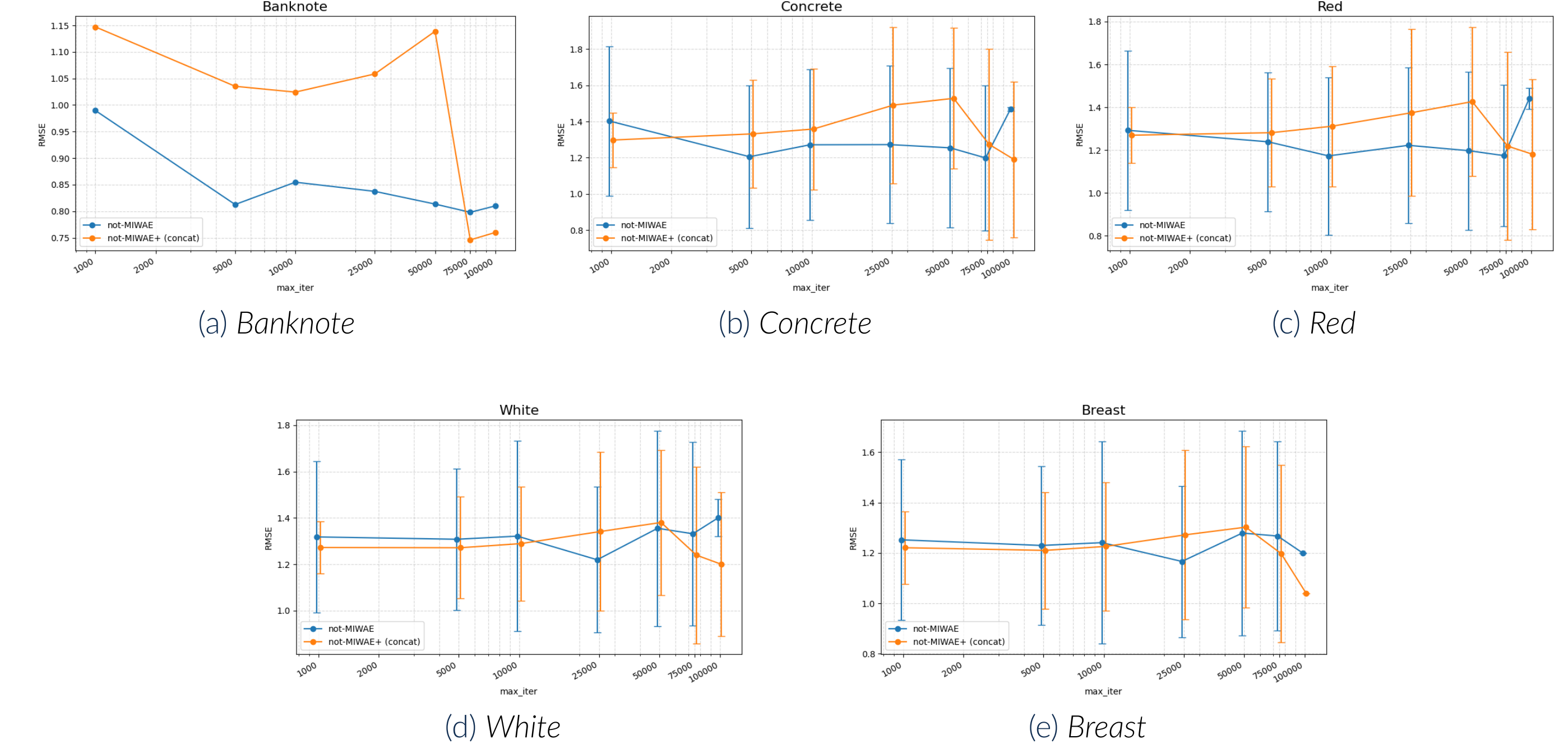


Figure 3. Imputation RMSE of two selected models as a function of training iterations.

Discussion

Main contribution: Successfully extended not-MIWAE for correlated missingness. Latent regime u captures structured dependencies. Concatenation architecture is most effective

Limitations: Higher computational cost, imputation performance is highly sensitive to training iterations, architecture-dependent, poor imputation performance on MAR and MCAR datasets

Future directions: Larger real-world datasets, correlated feature space, test with other MNAR missing mechanisms, hierarchical latent structure